

Illuminated Drone Show Simulation

Mathematical Modeling and Implementation

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Introduction

- Developed a simulation of an illuminated drone show.
- Goal: multiple drones forming shapes, transitioning smoothly, tracking moving objects.
- Combines mathematics, programming, and visual artistry.

Project Overview

- ① **Static Formation:** Transform drones into handwritten "TEKLAKIL".
- ② **Dynamic Morphing:** Transition from name to "Happy New Year!".
- ③ **Tracking with Shape Preservation:** Follow a moving object while keeping formation.

Core Problem

Each drone has:

$$\mathbf{x}_i \in \mathbb{R}^2, \quad \mathbf{v}_i \in \mathbb{R}^2$$

Update positions over time to create smooth, coordinated motion.

Smoothstep Function

Key tool for smooth motion:

$$s(t) = t^2(3 - 2t), \quad t \in [0, 1]$$

- $s'(0) = s'(1) = 0 \rightarrow$ start/stop smoothly
- $s''(t)$ continuous $\rightarrow$ natural acceleration
- Ease-in/out behavior

Position Interpolation

For drone movement:

$$\mathbf{x}(t) = (1 - s(p(t)))\mathbf{x}_{\text{start}} + s(p(t))\mathbf{x}_{\text{target}}$$

- $p(t)$: progress parameter per frame
- Smooth path between start and target

Individual Speeds

Assigning random speeds for asynchronous motion:

$$\text{speed}_i \sim \text{Uniform}(0.003, 0.006)$$

- Creates wave-like effect
- More visually engaging than synchronized motion

Handwritten Formation

TEKLAKIL

Figure: Handwritten reference image

Image Processing Steps

- ① Load grayscale image
- ② Resize to 300×100 px
- ③ Threshold dark pixels (< 120)
- ④ Sample 450 points
- ⑤ Center formation at origin

TEKLAKIL Formation Code

```
1 # Load and sample handwritten name
2 img = Image.open("handwritten.png").convert("L")
3 img = img.resize((300,100))
4 img_np = np.array(img)
5 ys, xs = np.where(img_np < 120)
6 points = np.vstack([xs, ys]).T
7 points = points - np.mean(points, axis=0)
```

Initial Drone Distribution

$$x_i \sim \text{Uniform}(-400, 400), \quad y_i \sim \text{Uniform}(-200, 200)$$

- 450 drones
- Random seed for reproducibility

Results: TEKLAKIL Formation



TEKLAKIL

Figure: 450 drones forming "TEKLAKIL"

Challenge of Morphing

- Maintain one-to-one correspondence
- Smooth continuous motion
- Handle larger greeting size

Generating Greeting Image

```
1 try:
2     greeting_img = Image.open("happy_new_year.png").convert("L")
3 except FileNotFoundError:
4     greeting_img = Image.new("L", (700,200), color=255)
5     draw = ImageDraw.Draw(greeting_img)
6     font = ImageFont.truetype("arial.ttf", 60)
7     draw.text((10,80),"Happy New Year!",fill=0,font=font)
```

Transition Animation

$$\mathbf{x}_i(t) = (1 - t)\mathbf{x}_i^{\text{name}} + t\mathbf{x}_i^{\text{greeting}}$$

- Smoothstep applied to t
- 700 drones for high resolution
- 200 frames, 6 sec duration

Results: Happy New Year!

A formation of blue dots arranged in a stylized, handwritten font to spell out the words "Happy New Year!". The dots are small and closely packed to form the letters, with clear spacing between the words.

Figure: Clear, well-spaced "Happy New Year!" formation

Core Concept

$$\mathbf{x}_i(t) = \mathbf{r}_i(t) + \mathbf{b}(t)$$

- $\mathbf{r}_i(t)$: position in formation
- $\mathbf{b}(t)$: reference trajectory (moving ball)

Ball Trajectory (Diagonal Path)

```
1 start_pos = np.array([-350,-300])
2 end_pos = np.array([350,300])
3 for i in range(total_frames):
4     t = smooth_step(i/(total_frames-1))
5     ball_positions[i] = (1-t)*start_pos + t*end_pos
```

Tracking Visualization



Figure: Start of tracking

Tracking Visualization (Mid End)

Drone Show — TEKLAKIL → Happy New Year on Ball Path!

Happy New Year!

Happy New Year

Figure: Middle and end of tracking

Challenges

- ① Choosing the right number of drones
- ② Coordinate system confusion
- ③ Preventing drone overlap
- ④ Tuning animation parameters
- ⑤ File management and robustness

Image Processing Steps

- ① Load grayscale image
- ② Resize to 300×100 px
- ③ Threshold dark pixels (< 120)
- ④ Sample 450 points
- ⑤ Center formation at origin

Note: This approach won't work well with low contrast photos, as the threshold may fail to distinguish text from background.

Handwritten Formation

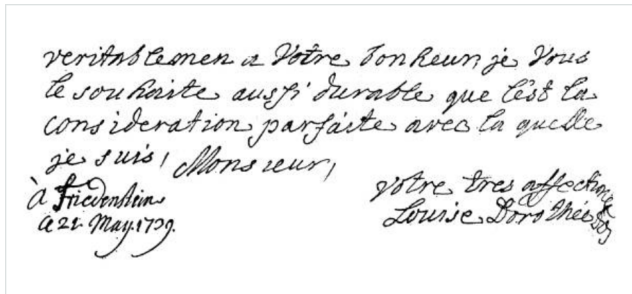


Figure: Handwritten reference image that won't work

Image Processing Limitation

Important Note

This thresholding approach won't work effectively with low contrast photos, where the difference between text and background is minimal.

Solution: Use high-contrast images with clear dark text on light backgrounds (or vice versa).

Key Lessons Learned

- **Technical:** Image processing, smooth interpolation, animation, parameter tuning
- **Design:** Simplicity, modularity, robustness, scalability
- **Mathematical:** Interpolation matters, superposition, asynchrony, continuity

Performance Metrics

Metric	Sub-prob 1	Sub-prob 2	Sub-prob 3
Drones	450	700	450
Frames	400	200	300
Duration	12s	6s	10s
FPS	20	20	20

Conclusion

- Successfully implemented all three sub-problems.
- Smooth, visually appealing drone show simulation.
- Learned important numerical, programming, and design principles.

Created with passion and mathematical precision - January 2026